Prodigy Anesthesia Quick Review

Hydrocephalus

What is hydrocephalus?	Hydrocephalus is an imbalance in the production and removal of cerebrospinal fluid that results in the accumulation of the fluid. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1000.*
What are the two types of hydrocephalus?	Communicating and obstructive. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1000.*
What is the pathway that CSF takes from the lateral ventricles where most of it is produced to the subarachnoid space?	CSF flows from the lateral ventricles into the third ventricle via the foramen of Monroe, through the aqueduct of Sylvius into the fourth ventricle and into the cerebromedullary cistern via the foramina of Lushke and Magendie. CSF enters the subarachnoid space from the cerebromedullary cistern and circulates around the brain and spinal cord before it is absorbed by the arachnoid granulations at the cerebral hemispheres. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 579.*
What is the rate of production of cerebrospinal fluid?	Cerebrospinal fluid is produced at a rate of 21 cc/hr by the choroid plexus and is absorbed by the arachnoid granulations over the cerebral hemispheres at a rate that maintains the total CSF volume at about 150 mL. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 579.*
How does obstructive hydrocephalus result in the accumulation of cerebrospinal fluid?	Obstructive hydrocephalus is caused by any obstruction of the pathways that CSF must follow to reach the arachnoid granulations. The obstruction may be direct, such as a clot within the passageway, or extrinsic compression of the passageway by a nearby tumor or other mass. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1000.*
How does communicating hydrocephalus result in the accumulation of cerebrospinal fluid?	Communicating hydrocephalus is a failure to absorb CSF. It is often due to abnormal function of the arachnoid granulations. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1000.*
What drugs can decrease CSF production?	Acetazolamide, isoflurane, sevoflurane, furosemide, corticosteroids, spironolactone, and vasoconstrictors all decrease CSF production. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 579.*

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How does isoflurane affect CSF volume?	Isoflurane has been shown to decrease CSF production which can result in a decreased volume. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 579.*
What anesthetic agent can increase CSF production?	Desflurane *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 581.*
What is normal-pressure hydrocephalus?	Normal pressure hydrocephalus is an increase in CSF that commonly occurs in the elderly. It may produce symptoms such as forgetfulness, memory difficulty, inattention, ataxia, and urinary incontinence. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 224.*
Into what location are ventricular shunts typically drained?	Ventricular shunts are most commonly drained into the peritoneal or pleural spaces. Less commonly, they may be drained into the atrial or choledochal spaces. An endoscopie third ventriculostomy (ETV) involves the placement of a burr hole into the lateral ventricle and then into the third ventricle. This allows the drainage of CSF into the cisterns below the third ventricle without the use of shunts. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 606.*
How is hydrocephalus usually treated?	Medical treatment may include the use of furosemide or acetazolamide to reduce CSF production. If the cause is obstruction by a mass or clot, the obstruction is removed. Otherwise, it is treated by placing a shunt into the ventricles to drain the abnormal accumulation of CSF. Lumbar punctures are used as a temporizing measure. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 606.*
What are the symptoms of hydrocephalus?	The symptoms depend on whether the problem is chronic or acute. Chronic hydrocephaly can present with vague symptoms such as headache and nausea. Acute hydrocephalus can produce lethargy, altered mentation, papilledema, decorticate or decerebrate posturing, bradycardia, hypertension, and ECG changes due to herniation of the brainstem. In infants, the cranium is often enlarged as the increased pressure expands the space between the open cranial plates. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 606.*

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What are the most common causes of congenital hydrocephalus?

The most common cause is obstruction of CSF flow due to anomalies such as Arnold-Chiari malformation, Dandy-Walker cysts, myelomeningocoele, aqueductal stenosis, arachnoid cysts, neoplasm, trauma-related defects, and vascular malformations.

Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 606.

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